

7.5 Finding the Area of Trapezoids, Parallelograms, and Rhombi

Lesson Introduction

 Benchmark	MA.7.GR.1.1 Apply formulas to find the areas of trapezoids, parallelograms, and rhombi.		 Learning Target	I can find the area of trapezoids, parallelograms, and rhombi.
 Benchmark Coherence	Building On MA.6.GR.2.2 Solve mathematical and real-world problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles.		Working Toward MA.912.GR.4.4 Solve mathematical and real-world problems involving the area of two-dimensional figures.	
 Essential Question	How do you find the areas of trapezoids, parallelograms, and rhombi in mathematical and real-world problems?			
 Lesson Narrative	Students find the areas of trapezoids, parallelograms, and rhombi using the formulas they developed in the previous lesson. The lesson prepares students for the next three lessons, where they will find the areas of composite figures.			
 Component Summary	7.5.1 Warm-Up (~10 min.)	Watch and reflect on video on theme park engineer career (<i>SE p. 322</i>)	7.5.3 Your Turn! (20 min.)	Solve real-world area problems (<i>SE p. 324</i>)
	7.5.2 Guided Practice (15 min.)	Recall and apply area formulas (<i>SE p. 323</i>)	7.5.4 Wrap-Up (~5 min.)	Identify, correct errors, and find area of rhombi (<i>SE p. 325</i>)
 Resources	Glossary	Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Parallelogram - Trapezoid - Rhombus	(Optional) Graph paper (Optional) Pattern blocks (Optional) Chart paper & markers		- Warm-Up 7.5.1 contains a video to accompany the question prompts. - Consider using data and/or observations from the previous and current lessons to determine small group intervention that can occur during 7.5.3 Your Turn!

 Teacher Tips	Common Misconceptions - Students may use the diagonal length instead of the perpendicular length for the height. - Students may neglect to use the order of operations when calculating the areas of trapezoids using the formula $A = \frac{1}{2}h(b_1 + b_2)$. - Students may not include the units with their answers or may not raise the units to the power of 2.	Things to Consider - Create and display an Anchor Chart with formulas, examples, and images. Show students how understanding the formulas prevents the need to memorize them. Include the various forms of the formulas that students derived in the previous lesson. - Consider allowing students to use calculators so they can focus on applying the formulas. - Consider having no-calculator questions where students turn their calculator over. Teachers should look for and consider opportunities for students to improve their fluency with rational numbers. For example, question 2 in 7.5.2 could be a no calculator problem but the questions in 7.5.3 could be calculator problems.
	Literacy and Language Routines Compare and Connect 7.5.2 Guided Practice provides an opportunity for students to think about and discuss the similarities and differences in the area formulas. Students also discuss how the formulas connect to the physical characteristics of the shapes.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.7.1: Real-World Contexts Students apply the area formulas for trapezoids, parallelograms, and rhombi to solve real-world problems in 7.5.3 Your Turn! Student interest and motivation will be ignited as they see the utility of the area formulas.
	 Differentiate Instruction Consider providing students with graph paper or pattern blocks so they can physically model the real-world parallelograms, trapezoids, and rhombi in 7.5.3 Your Turn! Alternatively, challenge students to a “Rhombus Scavenger Hunt” in which they must find and document as many different rhombi as possible as homework or an extension project if including dimensions and area.	
Lesson Notes:		

7.5.1 Warm-Up: A Day in the Life

Component Overview: Students begin by watching the career profile video on theme park engineers. After watching the 3:11 minute video, prompt students to independently reflect on the prompt. Consider having a few students share their responses with the whole class.



7.5.1 Warm-Up: A Day in the Life

Betsy McIver, a mechanical engineer, works in the Walt Disney Imagineering area and designs systems for the interactive attractions at the theme park. She mentions that her job can include creative left-brain thinking, as well as technical right-brain thinking.



She says that to be successful at her job, it is important to understand math, design, and theatre. She also mentions that many disciplines, such as designers and architects, all work together to make everything possible in the entertainment engineering industry.

1. Describe a ride you saw that looked exciting and entertaining.

Answers will vary. I saw a ride that went through Monsters Inc. that looked exciting.



Consider inviting a guest speaker, such as the technology teacher, an engineer, designer, or architect to join the class who can answer students' questions about their careers and the mathematics they use.

7.5.2 Guided Practice: Area of Parallelograms, Trapezoids, and Rhombi

Component Overview: Students practice calculating the areas of parallelograms, trapezoids, and rhombi. Consider using formative assessment data from the previous lesson to determine which students will be most successful working independently, with a pair, or in a small group that you assist. Monitor which students are using different strategies so you can ask them to share during the whole-class debrief.



7.5.2 Guided Practice: Area of Parallelograms, Trapezoids, and Rhombi

- Complete the table to match each type of quadrilateral with the formula to find its area.

$$A = \frac{1}{2}h(b_1 + b_2)$$

$$A = \frac{1}{2}d_1d_2$$

$$A = bh$$

Quadrilateral	Formula for the Area
parallelogram	$A = bh$
trapezoid	$A = \frac{1}{2}h(b_1 + b_2)$
rhombus	$A = \frac{1}{2}d_1d_2$

- The dimensions of three quadrilaterals are given in the table. Label each figure shown with the given dimensions in centimeters (cm). Then, calculate the area of each figure using the appropriate formula. Show your work.

Figure	Dimensions	Area
Parallelogram 	$b = 22 \text{ cm}$ $h = 8 \text{ cm}$	$A = bh$ $A = 22 \cdot 8$ $A = 176 \text{ cm}^2$



Compare and Connect

Provide students with an opportunity to think about and discuss the similarities and differences in the area formulas. Students can also discuss how the formulas connect to the physical characteristics of the shapes. Allow students to add equivalent formulas from the previous lesson that they prefer to use.

7.5.2 Guided Practice: Area of Parallelograms, Trapezoids, and Rhombi (continued)

Figure	Dimensions	Area
Trapezoid $h = \frac{1}{4} \text{ cm}$ $b_1 = \frac{7}{12} \text{ cm}$ $b_2 = \frac{11}{12} \text{ cm}$	$b_1 = \frac{7}{12} \text{ cm}$ $b_2 = \frac{11}{12} \text{ cm}$ $h = \frac{1}{4} \text{ cm}$	$A = \frac{1}{2} h(b_1 + b_2)$ $A = \frac{1}{2} \cdot \frac{1}{4} \cdot \left(\frac{7}{12} + \frac{11}{12} \right)$ $A = \frac{3}{16} \text{ cm}^2$
Rhombus $d_1 = 6.5 \text{ cm}$ $d_2 = 12.25 \text{ cm}$	$d_1 = 6.5 \text{ cm}$ $d_2 = 12.25 \text{ cm}$	$A = \frac{1}{2} d_1 d_2$ $A = \frac{1}{2} \cdot 6.5 \cdot 12.25$ $A = 39.8125 \text{ cm}^2$



The orientation of figures may be difficult for some students. Help students experience figures drawn with different orientations by purposely drawing figures differently.

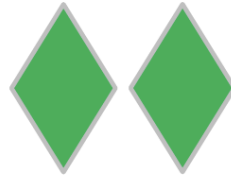
7.5.3 Your Turn!

Component Overview: Students practice finding the areas of parallelograms, trapezoids, and rhombi in a variety of real-world contexts. Students may engage with this activity independently or with a partner. Consider modifying the people, places, or things in the problems to contexts that appeal to your students' interests and everyday lives.



7.5.3 Your Turn!

- Alina is making rhombus-shaped earrings with green stones. The measure of one diagonal of each earring is $1\frac{3}{8}$ centimeters (cm), and the length of the other diagonal measures $1\frac{4}{9}$ cm. What is the area, in square centimeters (cm^2) of two earrings?



$$1\frac{71}{72}\text{cm}^2$$

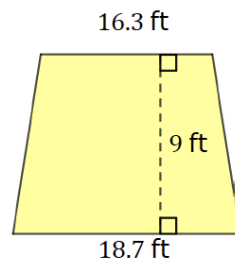


Consider pulling a small group during this activity to provide additional support to students based on observations from the previous and current lessons.



Does it matter which diagonal length you identify as d_1 and which you identify as d_2 ?

- One wall of Kel's apartment is shaped like a trapezoid with one base 18.7 feet (ft), the other base 16.3 ft, and a height of 9 ft.



Kel plans to paint this wall blue and needs to know how much paint to buy. What is the area of the wall Kel plans to paint?

$$157.5\text{ft}^2$$

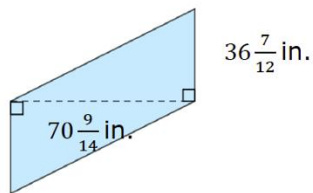


Why do you use square units when writing the area of rhombi, parallelograms, and trapezoids?

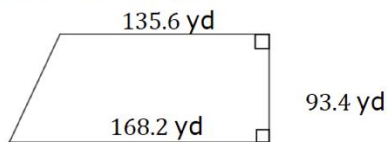
7.5.3 Your Turn! (continued)

3. Mika, a designer, is creating a geometric design in a tile floor. The design will be in the shape of a parallelogram with $36\frac{7}{12}$ inch (in.) bases and a height of $70\frac{9}{14}$ in. How many square inches of tile will she need for the design?

$$2,584\frac{59}{168}\text{ in}^2$$



4. The Smart family's horse pasture is in the shape of a trapezoid. One of the parallel sides is 168.2 yards (yd) long and the other is 135.6 yd long. The distance between these sides is 93.4 yd.



What is the area of the horse pasture?

$$14,187.46\text{ yd}^2$$

5. The Dockland Building in Hamburg, Germany is shaped like a parallelogram.

If the base of the building is 86 meters (m) and its height is 47 m, what is the area of the face of the building shown in the image?

$$4,042\text{ m}^2$$



In problems 3 and 5, does it matter which length you identify as the base and which you identify as the height?



Does it matter which base length you identify as b_1 and which you identify as b_2 ? Justify your response using numbers.



What does the question mean by "face" of the building? What shape is that?

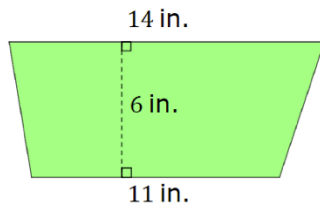
7.5.4 Wrap-Up: Ticket Out the Door

Component Overview: Students write a letter describing the correct and erroneous components of a sample of student work. Use data from this formative assessment to determine which students need remediation.



7.5.4 Wrap-Up: Ticket Out the Door

1. Don's work for finding the area of the trapezoid is shown below.



Don's Work
$A = \frac{1}{2} h(b_1 + b_2)$
$A = \frac{1}{2} (14)(11 + 6)$
$A = \frac{1}{2} (14)(17)$
$A = \frac{1}{2} \cdot 238$
$A = 119 \text{ in}^2$

- a. Write Don a note explaining what he did correctly and where he made an error.

Answers will vary. Don used the correct formula but he substituted the incorrect values for the base and the height and found the incorrect area.

- b. What is the area of the trapezoid?
75 in²



Students who struggle to express themselves in writing can circle the errors and then correct them or use an audio recorder (i.e. Flip Grid) or speech-to-text software.